

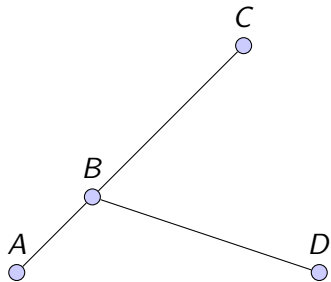
Graphs and its Applications

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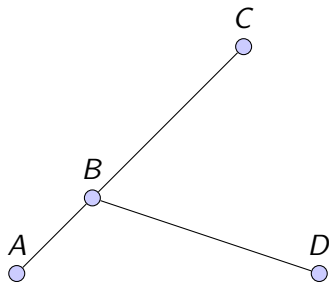
ICTS-TIFR

1 April 2025

Recap of Graphs

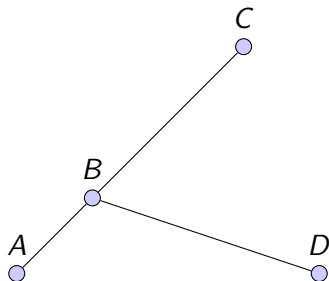


Recap of Graphs



- ▶ A, B, C, D are the **vertices**.
- ▶ AB, BC, BD are the **edges**.
- ▶ The number of edges that start at a given vertex is called the
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Question

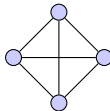
What are the degrees of the vertices A, B, C, D ?

Modeling Scenarios

4 people, all friends with each other

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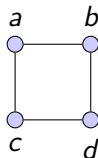


Modeling Scenarios

4 computers, each connected to 2 others

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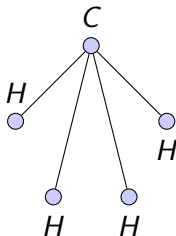


Modeling Scenarios

a molecule of methane, CH_4 , with 1 atom of carbon (C) connected to 4 atoms of hydrogen (H).

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Modeling problems

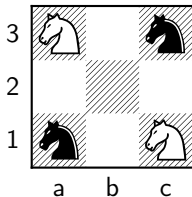
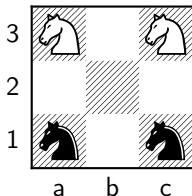
Can we create graphs to describe situations maths problems?

Modeling problems

There are 5 computers connected with several cables to make a network. Each cable connects a pair of computers and every computer is connected to two other computers.

Modeling problems

Several knights are situated on a 3×3 chessboard as shown in the Figure 1. Can they move, using the usual chess knight's move, to the position shown in Figure 2?

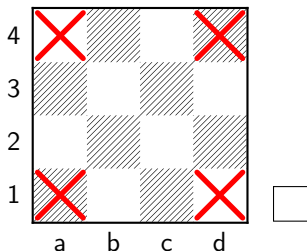


Modeling problems

A farmer has to take a tiger, a goat, and some melons across a river. Their boat has enough room for themselves plus either the tiger, the goat, or the melons. If they take the melons with them, the tiger will eat the goat. If they take the tiger across, the goat will eat the melons. Only when the farmer is present are the goat and the melons safe from their enemies. All the same, the farmer successfully carries the tiger, the goat, and the melons across the river. How?

Modeling problems

Can a knight travel around this board, pass through each square exactly once, and end on the same square it starts on?



Vertices and Edges

A couple invites 10 people to a house party. The hosts shake hands with everyone but each other. The guests shake hands with everyone else in the party. How many handshakes took place?

Vertices and Edges

IRTC has 9 computers. They want to link them into a network so that every computer would be connected with exactly 4 computers by network cables.

1. How many cables will they need? Can you draw such a configuration?
2. They decided to connect every computer with exactly 5 other computers. Is this a possible configuration?

Vertices and Edges

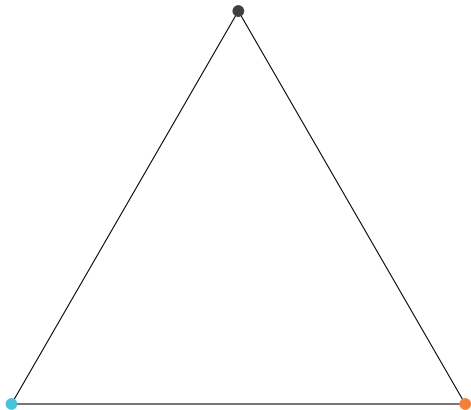
In Kottayam, there are 9 constituencies; some are connected by roads, and others are not. (A road connects a pair of constituencies with each other.) Is it possible that each of the constituencies is connected to exactly 3 other constituencies?

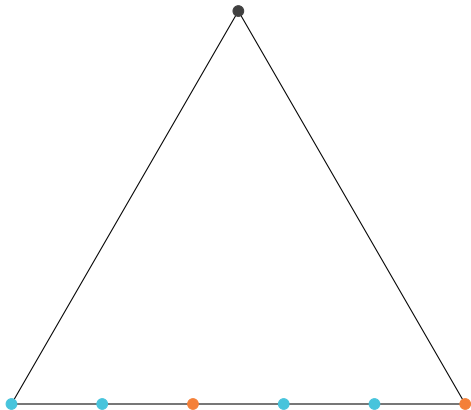
Sum of the degrees of all vertices should be an even $= 2$ times number of edges.

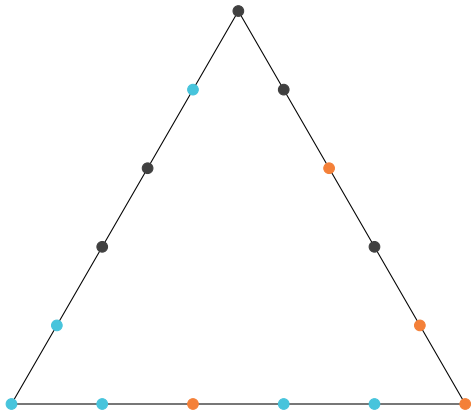
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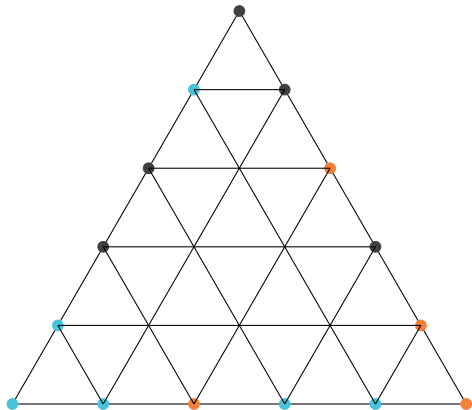
Handshake Theorem

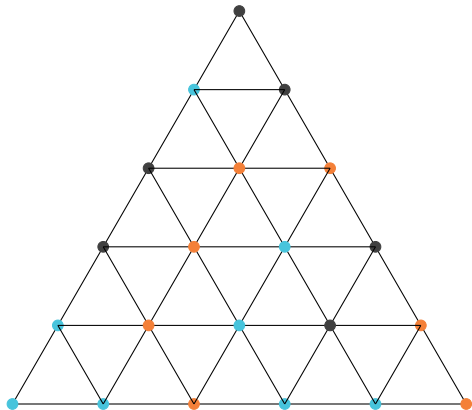
For any graph the number of vertices with an odd degree must be even.

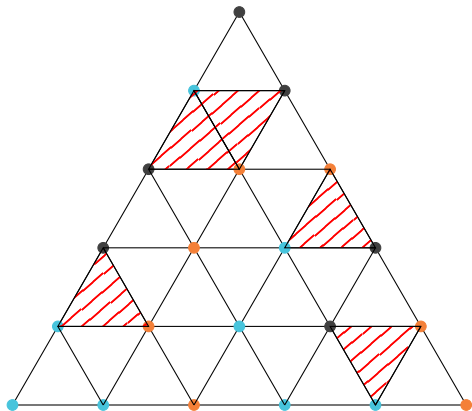


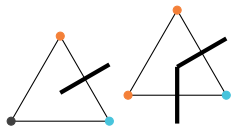
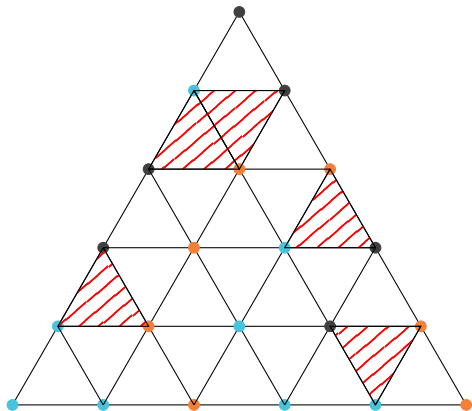


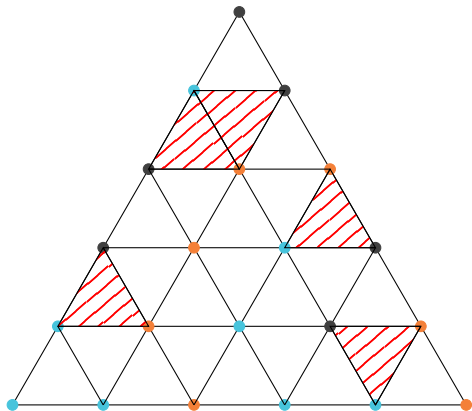


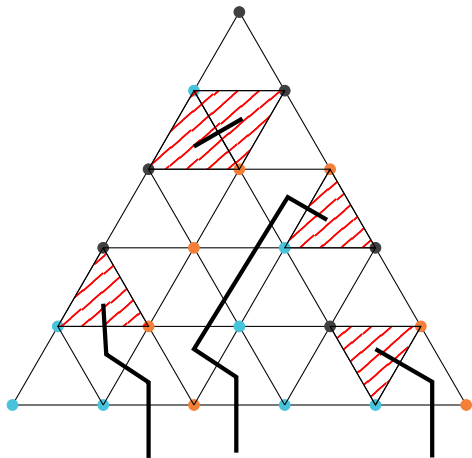


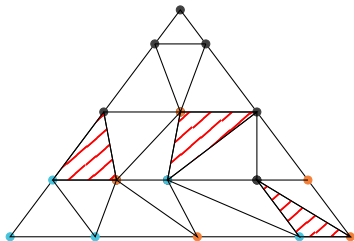


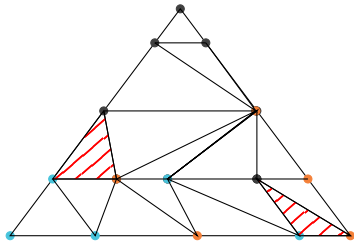












Sperner Coloring

1. Each of the three vertices A , B , and C of the initial triangle has a distinct color.
2. The vertices that lie along any edge of triangle ABC have only two colors, the two colors at the endpoints of the edge. For example, each vertex on AC must have the same color as A or C .

Sperner's Lemma

Sperner coloring of every triangulation has odd number of multicolored triangles

Coloring

In a group of 6 students, every pair of students are either friends or enemies. Prove that it is possible to choose 3 of these students so that they are either all friends or all enemies.

More Problems

The number of vertices with an odd degree must be even.
Graphs with odd numbers of odd degree vertices are impossible. Is the opposite equally true?

Can you draw a graph that has 5 vertices, in which the degrees of these vertices are 4, 4, 4, 2, and 2? Either draw such a graph or explain why it is impossible.

More Problems

The 8 heads of state of SAARC got together for a conference. According to a reporter, each of the heads of Afghanistan, Bangladesh, and Bhutan shook hands with 6 heads of state each, each of the heads of India, Maldives, and Nepal with 4, and each of the heads of Pakistan, and Sri Lanka with 1. Is the reporter correct?

More Problems

It is known that 3 edges meet at every vertex of a polyhedron. Is it possible for this polyhedron to have exactly 9 vertices?

More Problems

It is known that in a certain company, each employee has exactly 3 friends among the co-workers. Is it possible that this company has exactly 9 employees?



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